Name:_____ Period:_____

Entry/Exit Ticket: Sedimentary Rocks

Today we will be learning how sediments and dissolved minerals become sedimentary rocks so we can explain how the striped layers of Badlands National Park and the gypsum of White Sands formed.

Entry Ticket:

1. Sandstone is made of sand grains stuck together. How do you think loose sand becomes solid rock? Does this happen at Earth's surface or underground?

2. In today's demo, the cereal represents sediment. Predict what the melted marshmallow will represent.

During the Demo:

Word bank: sediment, deposition, compaction, cementation

Demo step	What I observed	What it represents (word bank)
Pouring layers of cereal into the pan		
Adding melted marshmallow		
Pressing the treats down with a spatula		

3. When cereal with no marshmallow was pressed, what happened? What does this show about why cementation is needed?

4. Use the model to describe how loose sediment becomes sedimentary rock. Use deposition, compaction, and cementation in order.

Exit Ticket:

5. Our model made a "rock" in 10 minutes, in one pan. Describe how real sedimentary rock forms differently in time (temporal scale) and in size (spatial scale). Do not use the ingredients as an answer.

Temporal: In our model, the rock formed in (how many minutes?) _____, but in real rock, it takes

(thousands or millions of what?) _____.

Spatial: In our model, the layers were (how thick? how big was the pan?) _____, but in real rock,

layers can be (how thick?) _____ and can cover (what kind of places?) _____

_____.

Success Criteria

I can describe the steps that turn sediment into sedimentary rock, from weathering to cementation.

I can explain how sediments settle into sorted layers and why fossils are found in sedimentary rock.